Sahaj Alatage

AI & Data Science undergraduate · desktop, mobile and AI software

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I build software end to end, from Rust and C++ internals to ML models and the interfaces on top. Shipped a Windows Dynamic Island, an offline PDF editor with 1,026 tests, a desktop AI assistant and a live forecasting system. Seeking software engineering internships.

EDUCATION

B.Tech, Artificial Intelligence & Data Science · Sanjay Ghodawat University (NIAT), Kolhapur — 2025 – 2029 · CGPA 8.42/10

Coursework: Data Structures, Database Management Systems, Building LLM Applications (A+), Introduction to Generative AI (O), Web Application Development.

EXPERIENCE

Intern · Antmark — React Native · TypeScript · C++

- Built **PIE**, an offline PDF and photo editor that edits existing text in the original font: shared C++ PDFium bridge (JNI), on-device OCR (ML Kit, Apple Vision), **1,026** Jest tests.
- Built **PhotoProof** end to end from Antmark's product brief: offline GPS-stamped photo reports, non-destructive edits, an ordered write queue and PDF export.

PROJECTS

FLOAT · Dynamic Island for Windows — Rust · Tauri 2 · React · GitHub

- Built one notch for media, notifications, clipboard history and quick settings on native Windows APIs (media controls, notification listener, WASAPI); released **v3.1** as an installer and MSIX.

HARIKA · AI assistant for the Windows desktop — Python · FastAPI · Tauri · React · GitHub

- Built an assistant with layered memory, a self-checking planner and permissioned tools; passes **280/280** scripted conversations across 8 capability areas.
- Made it model-agnostic (Groq, Gemini, Claude, or offline on Ollama) and shipped it as a Windows installer.

VayuCast · air-quality forecasting for 9 Indian cities — Python · XGBoost · FastAPI · Live · GitHub

- Built an end-to-end AQI forecaster: **±15.8** next-day error, **16–28%** better than a "same as today" baseline.
- Deployed a FastAPI service and Streamlit dashboard; GitHub Actions run daily forecasts, weekly retraining and **33** tests.

VERITY · financial reconciliation engine — Python · FastAPI · SQL · Live · GitHub

- Reconciled bank CSVs, invoices and chat screenshots into one audit trail; 18-table schema, **400+** tests, **96**-case benchmark.

Also built: ReconIQ (36 rules mapped to MITRE ATT&CK, 162 tests) · Aurelis Estates (GSAP site) · ResumePilot AI · this portfolio, a WebGL "OS" with live demos.

HACKATHONS & RECOGNITION

- **Finalist**, Makers Conclave 2026 (NIAT), team of 3: low-cost water monitoring with Wi-Fi sensors.
- **Meta Global AI Developer Hackathon 2026**: built MotherTongue Books, multi-agent AI that writes picture books natively in 8 languages, about $0.35 per book.
- **Razorpay AI Buildathon 2026**: VERITY · **PayPal AI Hackathon 2026**: Crumb, PayPal holds with AI buyer agents over MCP (in progress) · **Smart India Hackathon 2025**: participant.

SKILLS

Languages: Python, TypeScript, Rust, C++, Kotlin, SQL

Apps: React, Next.js, React Native, Expo, Tauri, WebGL

Backend & data: FastAPI, Pydantic, REST APIs, webhooks, PostgreSQL, Supabase

AI & ML: LLM applications, multi-agent pipelines, tool calling & MCP, scikit-learn, XGBoost, SHAP

Tools: Git, GitHub Actions, Docker, Vercel, pytest, Jest